

Ricky Nong

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Highly motivated & detail-oriented third-year Biomedical Engineering student with proven ability to work independently & collaboratively through accelerometer-based projects prioritizing electrical safety. Strong in oral & written communication skills through BMES VP Finance role & peer-reviewed publication. Eager to develop knowledge in power systems, SCADA automation & LAN Networks.

Education

Bachelor of Engineering, Biomedical Engineering

Sep. 2023 – present

Toronto Metropolitan University (Formerly Ryerson University) Toronto, ON

- **Cumulative GPA: 4.30 / 4.33**
- **Honours and Scholarships:** Dean's List - 2023/2024 — 2024/2025 — 2025/2026, TMU Entrance Scholarship - 4.15+ cGPA

Publication

Advancements in Piezoelectric Materials for Biomedical Applications: A Review

International Journal of Engineering Materials and Manufacture (IJEMM)

- Co-authored a peer-reviewed article regarding recent developments on piezoelectric biomaterials and their potential impact on tissue regeneration, cochlear implants, pacemakers, and more, while evaluating current limitations and future clinical potential.

Technical Skills

MATLAB — Simulink — C / C++ — Python — MultiSIM — MS Office — Quartus II — MPLAB X IDE — OnShape — Technical Documentation — Oscilloscope

Independent & Academic Projects

Accelerometer-Based Range of Motion Detector for Upper Limb Rehabilitation

Toronto Metropolitan University

Mar. 2026 - Apr. 2026
Toronto, ON

- Constructed and implemented a wearable accelerometer-based circuit with an Arduino to measure arm range of motion through graphical visualization and real-time angular measurement intended for physical rehabilitation, achieving accuracy within 3%.
- Soldered the accelerometer into an insulated plastic casing to isolate circuitry and improve electrical safety for patient protection.

Embedded Medical Fall Detection System for Elderly Care

Toronto Metropolitan University

Nov. 2025 - Dec. 2025

Toronto, ON

- Designed and implemented a wearable system with accelerometers to detect falls with 90% accuracy, implementing an embedded C program for real-time sensor data processing and immediate alert generation to enable faster response.
- Documented methods, results and analyses in a structured technical report, supporting compliance and knowledge sharing.
- Optimized sensor integration through collaboration, reducing false detections and improving overall system efficiency.

The Listener - Start Up Project

Mar. 2025 - present

- Collaborating within a multidisciplinary team to design an innovative device that converts sound energy to electrical energy.
- Prepared a professional pitch presentation highlighting its potential real-world impact & placed Top 5 in the Metropolitan Engineering Innovative Design Competition for idea, awarded \$5000 in funding to continue development of the project.
- Communicating with potential clients to assess interest and ideal use cases (e.g meeting net zero goals).

CAD Team Member - MARS Rocket Design Competition

Toronto Metropolitan University

Sep. 2024 - Jan. 2025

Toronto, ON

- Utilized the OnShape software for the modular design of rocket components, eventually constructing a functioning model
- Collaborated within a 5-student team to assemble a professional presentation emphasizing unique prototype design—allowing for an advance to the construction phase and subsequent rocket launch after approval from MARS competition judges.

Work Experience

Pharmacy Assistant

Oct. 2021 – Jan. 2022

Shoppers Drug Mart Mississauga, ON

- Performed accurate documentation of deliveries, new prescriptions, and vaccinations, following provincial health guidelines.
- Conducted refrigerated medication audits, ensuring inventory accuracy and compliance with proper storage conditions.
- Communicated effectively with patients to answer pharmaceutical inquiries in-store and over the phone.

Extracurriculars

Vice President of Finance

Jun. 2025 – present

Biomedical Engineering Society (BMES) – TMU Chapter Toronto, ON

- Prepared and managed BMES' annual budget proposal, securing funding for events held throughout the academic year.
- Documented, tracked and processed \$5000+ in reimbursement requests, ensuring accurate and timely financial compensation.
- Lead a team of directors to oversee event budget consulting, reimbursements and follow up to meet deadlines.
- Collaborated with BMES' cross-functional teams to plan events, discuss operations and maintain documentation of action items.
- Interviewed applicants for BMES' 2026-27 executive cohort, evaluating leadership & communication skills, and organizational fit.